# LEC4 — A tour of the Solar System



*Terrestrial planets are primarily rocky, while gas giant planets are primarily made of gasses

**Solar System:** A star (or multiple stars) orbited by planets, moon, and debris (asteroids, comets, and dust)

## Terrestrial Planets

- "Terrestrial" ⇒ Earth-like
- Composed mainly of rock (heavy metals)
- Small
- Thin or no atmosphere
- Few moons

VS.

## Gas Giant Planets

- "Jovian" ⇒ Jupiter-like
- Composed of gas and liquid with a small rocky core
- Rings
- Large
- Many moons

### Mercury

- No atmosphere
- Highly cratered surface
- No moons
- Rocky Exterior
- Huge Iron Core
- Temps range from −170°C to 425°C
- Has evidence of past volcanic activity

### Venus

- Thick (relative to earth) $CO_2$ atmosphere
- Lots of possibly active volcanos
- No moons
- About same size as earth
- Temps ~460°C
- Rotates very slowly

### Earth

- Wet planet with life
- Atmosphere of Nitrogen & Oxygen
- Actively ongoing volcanism
- Unusually big moon
- Only currently habitable planet in the solar system

### Mars

- Very thin $CO_2$ atmosphere
- Huge but extinct volcanoes
- Sub-surface and evidence for liquid water on surface in the past
- Several rovers sent

### Asteroid Belt

- In between Mars & Jupiter
- Millions of rocks & rubble piles
- Lots of space between them (1M km)
- Jupiters gravity prevents these from condensing into another rocky planet
- Asteroids are not spherical

### Jupiter

- Largest and most massive planet in solar system
- Thick gaseous atmosphere of hydrogen surrounding liquid hydrogen interior
- Possible rocky core
- Many moons (4 large, 80 total)
- Faint rings

- Read each blue section for important information on each celestial object

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## Saturn
- Composed of gas/light elements (hydrogen & helium)
- Rocky core
- Spectacular rings
- Many moons (60+)

## Uranus
- Composed of ice as well as gas
  → Ice giant
- Blue color due to methane in atmosphere
- Small solid core
- Many moons
- Faint rings
- Rotates on its side - axis tilt near 90°

## Neptune
- Composed of ice as well as gas
  → Ice giant
- Blue color due to methane in atmosphere
- Small solid core
- Many moons
- Faint rings
- Similar to Uranus

## Trans-Neptunian Objects / Dwarf Planet
- Many dwarf planets (largest is pluto)
- Dwarf planets are small icy bodies
- Largest known TNOs
  → Eris → Pluto → Makemake → Haumea
  → Sedna → Orcus → Quaoar → Varuna

## Kupiter Belt
- Ring/Disc of comets orbiting beyond Neptune
- Leftover material from formation of solar system

## Oort Cloud
- Spherical object of about 1 trillion comets out to 50,000AU

**Gravitational Perturbation:** A deviation in motion of a celestial object caused either by the gravitational force of a passing object or a collision with it

## Comets
- Rubble piles with ice

## "Oumuamua"
- Originated in another Solar System
- Ejected by a gravitational perturbation
  → Possibly from a planet in that system
- Flew through our Solar System
  → Not bound to Sun

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